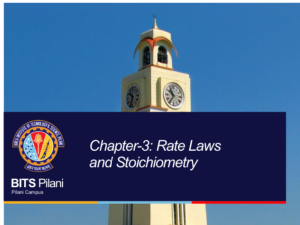




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Outline

- **PART 1: Rate Laws**
 - Relative Rates of Reaction
 - Reaction Order & Rate Law
 - Reaction Rate Constant, k
- **PART 2: Stoichiometry**
 - Batch System Stoichiometric Table
 - Flow System Stoichiometric Table
 - Calculation for Concentration in terms of Conversion

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Reactions

➤ **Molecularity:** The no. of atoms, ions or molecules involved in a reaction

- **Unimolecular Reaction:**
$$_mP^{2H} \rightarrow _mTh^{2H} + _mHe^d$$
- **Bimolecular Reaction**
$$Br \cdot + C_2H_6 \rightarrow HBr + C_2H_5 \cdot$$
- **Trimolecular reaction:**
$$2 NO + Cl_2 \rightarrow 2 NOCl$$

Most instances the reaction pathway follows a series of bimolecular reaction
$$2 NO + O_3 \rightarrow 2 NO_2$$

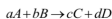
There is no known reactions with molecularity is 4

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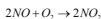
Relative Rates of Reaction



$$\frac{r_A}{-a} = \frac{r_B}{-b} = \frac{r_C}{c} = \frac{r_D}{d}$$

Reaction
Stoichiometry

EXAMPLE



If NO_2 formed at $4 \text{ mol/m}^3/\text{s}$
($r_{NO_2} = 4 \text{ mol/m}^3/\text{s}$), what is
the rate of formation of NO ?

$$\frac{r_{NO}}{-2} = \frac{r_{O_2}}{-1} = \frac{r_{NO_2}}{2}$$

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EXERCISE

- The Reaction:



is carried out in a reactor. If at a particular point,
the rate of disappearance of A is $10 \text{ mol/dm}^3/\text{s}$,
what are the rates of B and C?

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$$\frac{r_A}{-2} = \frac{r_B}{-3} = \frac{r_C}{5}$$

- Given, the rate of disappearance of A, $-r_A$, is $10 \text{ mol/dm}^3/\text{s}$
- Thus, solving the rates of B & C;

$$\frac{r_A}{-2} = \frac{r_B}{-3}$$

$$\frac{r_A}{-2} = \frac{r_C}{5}$$

$$r_B = \frac{-3}{-2} (-10 \text{ mol/dm}^3/\text{s})$$

$$r_C = \frac{5}{-2} (-10 \text{ mol/dm}^3/\text{s})$$

$$= -15 \text{ mol/dm}^3/\text{s}$$

$$= 25 \text{ mol/dm}^3/\text{s}$$

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Powerlaw Kinetics

- Consider a reaction $\sum \nu_j A_j = 0$, taking place in a **closed, isothermal, constant pressure** system

$$r = k C_A^{\alpha} C_B^{\beta} \dots C_N^{\gamma} = k \prod_{j=1}^N C_j^{\nu_j}$$

$$-r_A = k C_A^{\alpha} C_B^{\beta}$$

The rxn is α order wrt reactant A

AND

The rxn is β order wrt reactant B

The overall order of the reaction, n ; $n = \alpha + \beta$

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- The unit of the specific reaction, k , will vary with the order of reaction.



$$k = \frac{(\text{Concentration})^{1-n}}{\text{Time}}$$

Zero order ($n=0$)	$-r_A = k_A$	$\{k\} = \text{mol} / \text{dm}^3 \cdot \text{s}$
First order ($n=1$)	$-r_A = k_A C_A$	$\{k\} = \text{s}^{-1}$
Second order ($n=2$)	$-r_A = k_A C_A^2$	$\{k\} = \text{dm}^3 / \text{mol} \cdot \text{s}$
Third order ($n=3$)	$-r_A = k_A C_A^3$	$\{k\} = (\text{dm}^3 / \text{mol})^2 \cdot \text{s}^{-1}$

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Law of mass action

- Consider a reaction $\sum \nu_j A_j = 0$, taking place in a **closed, isothermal, constant pressure** system

$$r = k \prod_{j=1}^N C_j^{\nu_j}$$

$$q_j = \frac{1}{2} (|v_j| - v_j)$$

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Reversible reaction

- Consider a reaction $\sum \nu_j A_j = 0$, taking place in a **closed, isothermal, constant pressure** system

$$r = k_f \prod_{j=1}^N C_j^{\nu_j} - k_b \prod_{j=1}^N C_j^{\nu_j}$$

$$q_j = \frac{1}{2} (|v_j| - v_j)$$

$$q_j = \frac{1}{2} (|v_j| + v_j)$$

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Reaction Order & Rate Law

- The reaction rate (rate of disappearance) depends on temperature and composition.

- It can be written as the product of **reaction rate constant, k_A** and a function of **concentrations (activities) of the reactants** involved in the reaction:

$$-r_A = [k_A(T)] f(C_A, C_B, \dots)$$

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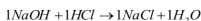
Rate law is a kinetic expression that gives the relationship between reaction rate, r_A , and concentration.

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- For reaction in which the stoichiometric coefficient is 1 for ALL species:



we shall delete the subscript on the specific reaction rate, (e.g.: A in k_A) to let

$$k = k_{\text{NaOH}} = k_{\text{HCl}} = k_{\text{NaCl}} = k_{\text{H}_2\text{O}}$$

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POWER LAW MODELS & ELEMENTARY RATE LAWS

Elementary reaction: a chemical reaction in which one or more of the chemical species react directly to form products in a **single reaction step** and with a single transition state.

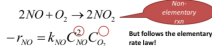


Unimolecular reaction

Bimolecular reaction

Elementary rate law:

The rxn is said to **follow the elementary rate law** if the stoichiometric coefficients are **IDENTICAL** to the reaction order of each species.



Non elementary rxn

But follows the elementary rate law!

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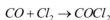
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NON-ELEMENTARY RATE LAWS

- Non-elementary rate laws: reactions that **do not follow** simple rate laws (power rate laws).
- Example 1: Homogeneous Rxn



Gas phase synthesis of phosgene

The kinetic rate law is:

$$-r_{\text{CO}} = k C_{\text{CO}} C_{\text{Cl}_2}^{1.5}$$

Rxn order: first order wrt to CO, three-halves order wrt Cl_2 , five-halves order overall.

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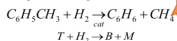
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NON-ELEMENTARY RATE LAWS

- Example 2: Heterogeneous Rxn



Gas-solid catalyzed rxn. Hydrodemethylation of toluene (T)

The rate of disappearance of toluene per mass of catalyst is:

$$-r_T' = \frac{k P_{\text{H}_2} P_{\text{T}}}{1 + K_B P_{\text{B}} + K_T P_{\text{T}}}$$

In terms of partial pressure rather than concentrations

where K_B & K_T is the adsorption constants.

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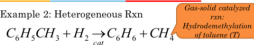
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NON-ELEMENTARY RATE LAWS

- Example 2: Heterogeneous Rxn



The rate of disappearance of toluene per mass of catalyst is:

$$-r_T' = \frac{kP_H P_T}{1 + K_B P_B + K_T P_T}$$

In terms of partial pressure rather than concentrations

where K_B & K_T is the adsorption constants.

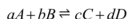
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Reversible reaction



- For reversible rxn, all rate laws *must* reduce to the **thermodynamic** relationship relating the reacting species concentrations at equilibrium.

$$K_C = \frac{C_C^c C_D^d}{C_A^a C_B^b}$$

Thermodynamic Equilibrium Relationship

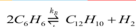
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REVERSIBLE REACTIONS

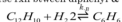


symbolically; $2B \xrightleftharpoons[k_B]{k_B} D + H_2$

The rate of disappearance of benzene;

$$-r_{B,forward} = k_B C_B^2 \quad \text{OR} \quad r_{B,forward} = -k_B C_B^2$$

The reverse rxn between diphenyl & hydrogen;



The rate of formation of benzene (in reverse direction);

$$r_{B,reverse} = k_{-B} C_D C_{H_2}$$

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REVERSIBLE REACTIONS

The net rate of formation of benzene is;

$$\begin{aligned} r_B &\equiv r_{B,net} = r_{B,forward} + r_{B,reverse} \\ &= -k_B C_B^2 + k_{-B} C_D C_{H_2} \end{aligned}$$

Multiplying both sides by -1, we obtain the rate law of disappearance of benzene, $-r_B$

$$\begin{aligned} -r_B &= k_B C_B^2 - k_{-B} C_D C_{H_2} \\ &= k_B \left(C_B^2 - \frac{k_{-B}}{k_B} C_D C_{H_2} \right) \end{aligned}$$

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REVERSIBLE REACTIONS

$$-r_B = k_B \left(C_B^2 - \frac{k_{-B}}{k_B} C_D C_{H_2} \right)$$

Replacing the ratio of the reverse & forward rate law constant by equilibrium constants;

$$-r_B = k_B \left(C_B^2 - \frac{C_D C_{H_2}}{K_C} \right)$$

where

$$\frac{k_B}{k_{-B}} = K_C = \text{Concentration equilibrium constant}$$

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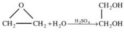
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Examples

A. First-Order Rate Laws

- (1) $C_2H_6 \longrightarrow C_2H_4 + H_2$ $-r_A = kC_{C_2H_6}$
- (2) $gN \rightarrow nCl \longrightarrow gCl + N_2$ $-r_A = kC_{gN} + nC_1$
- (3)  $-r_A = kC_{(CH_3)_2O}C_{H_2O}$
- (4) $CH_3COCH_3 \longrightarrow CH_3CO + CH_4$ $-r_A = kC_{(CH_3)_2CO}$
- (5) $nC_4H_{10} \rightleftharpoons iC_4H_{10}$ $-r_A = k[C_{nC_4} - C_{iC_4}/K_C]$

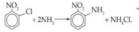
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B. Second-Order Rate Laws

- (1)  $-r_A = k_{ONCl} C_{ONCl} C_{NH_3}$
- (2) $CNBr + CH_3NH_2 \longrightarrow CH_3Br + NCNH_2$ $-r_A = kC_{CNBr}C_{CH_3NH_2}$
- (3) $CH_3COOC_2H_5 + C_4H_9OH \rightleftharpoons CH_3COOC_4H_9 + C_2H_5OH$
A + B $\rightleftharpoons$ C + D $-r_A = k[C_A C_B - C_C C_D / K_C]$

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C. Non elementary Rate Laws

- (1) $CH_3CHO \longrightarrow CH_4 + CO$ $-r_{CH_3CHO} = kC_{CH_3CHO}^2$
- (2)  $-r_A = \frac{k_1 P_1 \cdot P_2 P_3 \cdot k_2}{1 + k_3 P_2 + k_4 P_3}$

Chlorine (2) $\longrightarrow$ Benzene (3) + Benzene (2)

D. Enzymatic Reactions (Urea (U) + Urease (E))



$$-r_A = \frac{kC_U}{K_m + C_U}$$

E. Bioman Reactions



$$-r_A = \frac{k_1 C_S C_C}{K_m + C_S}$$

Note: The rate constant, k , and activation energies for a number of the reactions in these reactions are given in the Data Base on the CD-ROM and Summary Notes.

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THE REACTION RATE CONSTANT

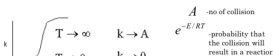
$$k_A(T) = Ae^{-E/RT}$$

Arrhenius equation

A = pre-exponential factor or frequency factor
 E = activation energy, J/mol or cal/mol
 R = gas constant = 8.314 J/mol-K = 1.987 cal/mol-K
 T = absolute temperature, K



Svante Arrhenius (1869–1927), Nobel Prize in Chemistry, 1903



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THE REACTION RATE CONSTANT

$$k_A(T) = Ae^{-E/RT}$$

- Activation energy is a measure of the **minimum energy** that the **reacting molecules must have** in order for the reaction to occur (energy required to reach transition state).



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THE REACTION RATE CONSTANT

$$k_A(T) = Ae^{-E/RT}$$

Taking a natural logarithm;

$$\ln k_A = \ln A - \frac{E}{R} \left(\frac{1}{T} \right)$$

- E !, k !, $-r$!

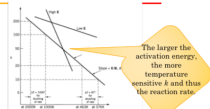
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Arrhenius plot



- Reactions with high activation energies are very temperature-sensitive; reactions with low activation energies are relatively temperature-insensitive.
- Any given reaction is much more temperature-sensitive at a low temperature than at a high temperature.

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BATCH SYSTEMS STOICHIOMETRIC TABLE

- Purpose of developing stoichiometric table:
 - To determine the no of moles of each species remaining at a conversion of X .

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BATCH SYSTEMS STOICHIOMETRIC TABLE

- Components of stoichiometric table:

refers to moles of species reacted or formed

Species	Initially (mol)	Change (mol)	Remaining (mol)
A			
B			
C			
D			
Inerts(I)			
Totals			

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BATCH SYSTEMS STOICHIOMETRIC TABLE



- Recall from Chapter 2:

$$X = \frac{(N_{A0} - N_A)}{N_{A0}}$$

$$N_A = N_{A0} - N_{A0}X$$

moles of A reacted

- Factorizing:

$$N_A = N_{A0}(1 - X)$$

moles of A remaining in the reactor at a conversion of X

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BATCH SYSTEMS STOICHIOMETRIC TABLE

$$\text{Moles B reacted, } N_B = \frac{\text{Moles B reacted}}{\text{Moles A reacted}} (\text{Moles A reacted})$$

$$= \frac{b}{a} (N_{A0}X)$$

$$\text{Moles C formed, } N_C = \frac{c}{a} (N_{A0}X)$$

$$\text{Moles D formed, } N_D = \frac{d}{a} (N_{A0}X)$$

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BATCH SYSTEMS STOICHIOMETRIC TABLE

$$\text{moles B remaining} = N_{B0} - \frac{b}{a} N_{A0} X$$

in the system, N_B

↓ moles of B initially in the system

↓ moles of B reacted

$$N_C = N_{C0} + \frac{c}{a} N_{A0} X$$

↓ moles of C formed

$$N_D = N_{D0} + \frac{d}{a} N_{A0} X$$

↓ moles of D formed

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BATCH SYSTEMS STOICHIOMETRIC TABLE

Species	Initially (mol)	Change (mol)	Remaining (mol)
A	N_{A0}	$- (N_{A0} X)$	$N_A = N_{A0} - N_{A0} X$
B	N_{B0}	$- \frac{b}{a} (N_{A0} X)$	$N_B = N_{B0} - \frac{b}{a} N_{A0} X$
C	N_{C0}	$+ \frac{c}{a} (N_{A0} X)$	$N_C = N_{C0} + \frac{c}{a} N_{A0} X$
D	N_{D0}	$+ \frac{d}{a} (N_{A0} X)$	$N_D = N_{D0} + \frac{d}{a} N_{A0} X$
I	N_{I0}	-	$N_I = N_{I0}$
Totals	N_{T0}	-	$N_T = N_{T0} + \left(\frac{d}{a} + \frac{c}{a} - \frac{b}{a} - 1 \right) N_{A0} X$

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BATCH SYSTEMS STOICHIOMETRIC TABLE

- o Total no of moles per mole of A reacted can be calculated as:

$$N_T = N_{T0} + \left(\frac{d}{a} + \frac{c}{a} - \frac{b}{a} - 1 \right) N_{A0} X$$

$$= N_{T0} + \delta N_{A0} X$$

where

Change in the total number of moles per mole of A reacted

$$\delta = \frac{d}{a} + \frac{c}{a} - \frac{b}{a} - 1$$

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BATCH SYSTEMS STOICHIOMETRIC TABLE

$$-r_A = k C_A \quad -r_A = k C_A^2 \quad -r_A = k_A C_A^3$$

Can we express concentration of each species??

Species	Initially	Change	Remaining	Concentration
A				
B				
C				
D				
I				
Totals				

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BATCH SYSTEMS STOICHIOMETRIC TABLE

o **Concentration** of each species in terms of conversion can be expressed as:

$$C_A = \frac{N_A}{V} = \frac{N_{A0}(1-X)}{V}$$

$$C_B = \frac{N_B}{V} = \frac{N_{B0} - (b/a)N_{A0}X}{V}$$

$$C_C = \frac{N_C}{V} = \frac{N_{C0} + (c/a)N_{A0}X}{V}$$

$$C_D = \frac{N_D}{V} = \frac{N_{D0} + (d/a)N_{A0}X}{V}$$

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Recall from stoichiometry table:

Remaining (mol)	
A	$N_A = N_{A0} - N_{A0}X$
B	$N_B = N_{B0} - \frac{b}{a}N_{A0}X$
C	$N_C = N_{C0} + \frac{c}{a}N_{A0}X$
D	$N_D = N_{D0} + \frac{d}{a}N_{A0}X$

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BATCH SYSTEMS STOICHIOMETRIC TABLE

$$C_B = \frac{N_{B0} - (b/a)N_{A0}X}{V}$$

$$= \frac{N_{A0} \left[\frac{N_{B0}}{N_{A0}} - (b/a)X \right]}{V} = \frac{N_{A0} [\Theta_B - (b/a)X]}{V}$$

$$C_C = \frac{N_{C0} + (c/a)N_{A0}X}{V}$$

$$= \frac{N_{A0} \left[\frac{N_{C0}}{N_{A0}} + (c/a)X \right]}{V} = \frac{N_{A0} [\Theta_C + (c/a)X]}{V}$$

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BATCH SYSTEMS STOICHIOMETRIC TABLE

$$C_D = \frac{N_{D0} + (d/a)N_{A0}X}{V}$$

$$= \frac{N_{A0} \left[\frac{N_{D0}}{N_{A0}} + (d/a)X \right]}{V}$$

$$= \frac{N_{A0} [\Theta_D + (d/a)X]}{V}$$

$$\Theta_i = \frac{N_{i0}}{N_{A0}} = \frac{C_{i0}}{C_{A0}} = \frac{y_{i0}}{y_{A0}}$$

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BATCH SYSTEMS STOICHIOMETRIC TABLE

Species	Initial	Change	Remaining	Concentration
A	N_{A0}	$-(N_{A0}X)$	$N_A = N_{A0} - N_{A0}X$	$C_A = \frac{N_{A0}(1-X)}{V}$
B	N_{B0}	$-\frac{b}{a}(N_{A0}X)$	$N_B = N_{B0} - \frac{b}{a}N_{A0}X$	$C_B = \frac{N_{A0}[\Theta_B - (b/a)X]}{V}$
C	N_{C0}	$+\frac{c}{a}(N_{A0}X)$	$N_C = N_{C0} + \frac{c}{a}N_{A0}X$	$C_C = \frac{N_{A0}[\Theta_C + (c/a)X]}{V}$
D	N_{D0}	$+\frac{d}{a}(N_{A0}X)$	$N_D = N_{D0} + \frac{d}{a}N_{A0}X$	$C_D = \frac{N_{A0}[\Theta_D + (d/a)X]}{V}$
I	N_{I0}	-	$N_I = N_{I0}$	C_{I0}

$$N_B = N_{B0} - \frac{b}{a}N_{A0}X = N_{A0} \left(\frac{N_{B0}}{N_{A0}} - \frac{b}{a}X \right)$$

$$= N_{A0} \left(\Theta_B - \frac{b}{a}X \right)$$

$$\Theta_i = \frac{N_{i0}}{N_{A0}} = \frac{C_{i0}}{C_{A0}} = \frac{y_{i0}}{y_{A0}}$$

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BATCH SYSTEMS STOICHIOMETRIC TABLE

Species	Initially	Change	Remaining	Concentration
A	N_{A0}	$-(N_{A0}X)$	$N_A = N_{A0} - N_{A0}X$	$C_A = \frac{N_{A0}(1-X)}{V}$
B	N_{B0}	$-\frac{b}{a}(N_{A0}X)$	$N_B = N_{B0} \left(\Theta_B - \frac{b}{a}X \right)$	$C_B = \frac{N_{B0} \left[\Theta_B - (b/a)X \right]}{V}$
C	N_{C0}	$\frac{c}{a}(N_{A0}X)$	$N_C = N_{C0} \left(\Theta_C + \frac{c}{a}X \right)$	$C_C = \frac{N_{C0} \left[\Theta_C + (c/a)X \right]}{V}$
D	N_{D0}	$\frac{d}{a}(N_{A0}X)$	$N_D = N_{D0} \left(\Theta_D + \frac{d}{a}X \right)$	$C_D = \frac{N_{D0} \left[\Theta_D + (d/a)X \right]}{V}$
I	N_{I0}	-	$N_I = N_{I0}$	C_{I0}

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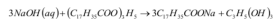
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BATCH SYSTEMS STOICHIOMETRIC TABLE

EXAMPLE

Given the saponification for the formation of soap from *aqueous* caustic soda & glyceryl stearate is:



Letting X the conversion of sodium hydroxide, set up a stoichiometric table **expressing the concentration of each species** in terms of its initial concentration and the conversion.

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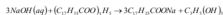
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BATCH SYSTEMS STOICHIOMETRIC TABLE

We know that this is a liquid-phase reaction.

Therefore, $V=V_0$



$$C_A = \frac{N_{A0}(1-X)}{V} = \frac{N_{A0}(1-X)}{V_0} = C_{A0}(1-X)$$

$$C_B = \frac{N_{B0} \left[\Theta_B - (b/a)X \right]}{V_0} = C_{B0} \left[\Theta_B - \frac{1}{3}X \right]$$

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BATCH SYSTEMS STOICHIOMETRIC TABLE

Species	Initially	Change	Remaining	Concentration
A	N_{A0}	$-N_{A0}X$	$N_A = N_{A0}(1-X)$	$C_A = C_{A0}(1-X)$
B	N_{B0}	$-\frac{1}{3}(N_{A0}X)$	$N_B = N_{B0} \left(\Theta_B - \frac{1}{3}X \right)$	$C_B = C_{B0} \left(\Theta_B - \frac{1}{3}X \right)$
C	N_{C0}	$N_{A0}X$	$N_C = N_{C0}(\Theta_C + X)$	$C_C = C_{C0}(\Theta_C + X)$
D	N_{D0}	$\frac{1}{3}(N_{A0}X)$	$N_D = N_{D0} \left(\Theta_D + \frac{1}{3}X \right)$	$C_D = C_{D0} \left(\Theta_D + \frac{1}{3}X \right)$
I	N_{I0}	-	$N_I = N_{I0}$	C_{I0}
Total	N_{T0}	0	$N_T = N_{T0}$	

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FLOW SYSTEMS STOICHIOMETRIC TABLE

○ Purpose of developing stoichiometric table:

- To determine the effluent flow rate of each species at a conversion of X .



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FLOW SYSTEMS STOICHIOMETRIC TABLE

○ Components of stoichiometric table:

Species	Feed rate to reactor (mol/time)	Change within the reactor (mol/time)	Effluent rate from reactor (mol/time)
A			
B			
C			
D			
I			
Totals			

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FLOW SYSTEMS STOICHIOMETRIC TABLE

Species	Feed rate to reactor (mol/time)	Change within the reactor (mol/time)	Effluent rate from reactor (mol/time)	Concentration (mol/L)
A	F_{A0}	$-(F_{A0}X)$	$F_A = F_{A0} - F_{A0}X$	$C_A = \frac{F_{A0}(1-X)}{v}$
B	$F_{B0} = \Theta_B F_{A0}$	$-\frac{b}{a}(F_{A0}X)$	$F_B = F_{B0} \left(\Theta_B - \frac{b}{a}X \right)$	$C_B = \frac{F_{B0} \left[\Theta_B - (b/a)X \right]}{v}$
C	$F_{C0} = \Theta_C F_{A0}$	$\frac{c}{a}(F_{A0}X)$	$F_C = F_{C0} \left(\Theta_C + \frac{c}{a}X \right)$	$C_C = \frac{F_{C0} \left[\Theta_C + (c/a)X \right]}{v}$
D	$F_{D0} = \Theta_D F_{A0}$	$\frac{d}{a}(F_{A0}X)$	$F_D = F_{D0} \left(\Theta_D + \frac{d}{a}X \right)$	$C_D = \frac{F_{D0} \left[\Theta_D + (d/a)X \right]}{v}$
I	$F_{I0} = \Theta_I F_{A0}$	*	$F_I = F_{I0} \Theta_I$	$C_I = \frac{F_{I0} \Theta_I}{v}$
Totals	F_{T0}		$F_T = F_{T0} + \delta F_{A0} X$	

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Example

- Given a liquid phase reaction:
 $A + 2B \rightarrow C + D$

The initial concentration of A and B are 1.8 kmol/m³ and 6.6 kmol/m³ respectively. Construct a stoichiometric table for a flow system considering A as the basis of calculation.

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$$\Theta = \frac{F_{i0}}{F_{A0}} = \frac{C_{i0}}{C_{A0}} = \frac{y_{i0}}{y_{A0}}$$



Given:

$$C_{A0} = 1.8 \text{ kmol/m}^3$$

$$C_{C0} = 0 \text{ kmol/m}^3$$

Since C & D are products.

$$C_{B0} = 6.6 \text{ kmol/m}^3$$

$$C_{D0} = 0 \text{ kmol/m}^3$$

From stoichiometry, we know that,

$$a = 1 \quad b = 2 \quad c = 1 \quad d = 1$$

$$\delta = \frac{d}{a} + \frac{c}{a} - \frac{b}{a} = -1$$

$$\Theta_i = \frac{C_{i0}}{C_{A0}}$$

$$\Theta_B = \frac{6.6}{1.8} = 3.67$$

$$\Theta_C = \frac{0}{1.8} = 0$$

$$\Theta_D = \frac{0}{1.8} = 0$$

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Species	Feed rate to reactor (mol/time)	Change within the reactor (mol/time)	Effluent rate from reactor (mol/time)
A	F_{A0}	$-(F_{A0}X)$	$F_A = F_{A0}(1-X)$
B	$F_{B0} = F_{A0}\Theta_B$	$-2(F_{A0}X)$	$F_B = F_{A0}(\Theta_B - 2X)$
C	$F_{C0} = F_{A0}\Theta_C$	$(F_{A0}X)$	$F_C = F_{A0}X$
D	$F_{D0} = F_{A0}\Theta_D$	$(F_{A0}X)$	$F_D = F_{A0}X$
Totals	F_{T0}		$F_T = F_{T0} + \delta F_{A0}X$

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Substituting the numerical values;

Species	Feed rate to reactor (mol/time)	Change within the reactor (mol/time)	Effluent rate from reactor (mol/time)
A	F_{A0}	$-F_{A0}X$	$F_A = F_{A0}(1-X)$
B	$F_{B0} = 3.67 F_{A0}$	$-2F_{A0}X$	$F_B = F_{A0}(3.67 - 2X)$
C	$F_{C0} = 0$	$F_{A0}X$	$F_C = F_{A0}X$
D	$F_{D0} = 0$	$F_{A0}X$	$F_D = F_{A0}X$
Totals	F_{T0}		$F_T = F_{T0} - F_{A0}X$

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CONCENTRATION IN TERMS OF CONVERSION

For liquid phase:

$$\bullet \text{ Batch System: } V = V_0$$

$$C_A = \frac{N_A}{V} = \frac{N_{A0}(1-X)}{V}$$

$$C_B = \frac{N_B}{V} = \frac{N_{B0}[\Theta_B - (b/a)X]}{V} = \frac{N_{B0}[\Theta_B - (b/a)X]}{V_0} = C_{A0}[\Theta_B - (b/a)X]$$

$$C_C = \frac{N_C}{V} = \frac{N_{C0}[\Theta_C + (c/a)X]}{V} = \frac{N_{C0}[\Theta_C + (c/a)X]}{V_0} = C_{A0}[\Theta_C + (c/a)X]$$

$$C_D = \frac{N_D}{V} = \frac{N_{D0}[\Theta_D + (d/a)X]}{V} = \frac{N_{D0}[\Theta_D + (d/a)X]}{V_0} = C_{A0}[\Theta_D + (d/a)X]$$

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CONCENTRATION IN TERMS OF CONVERSION

For liquid phase:

- Flow System - $U = U_0$

$$C_A = \frac{F_A}{U} = \frac{F_{A0}(1-X)}{U}$$

$$C_B = \frac{F_B}{U} = \frac{F_{B0}[\Theta_B - (b/a)X]}{U} = \frac{F_{B0}[\Theta_B - (b/a)X]}{U_0} = C_{B0}[\Theta_B - (b/a)X]$$

$$C_C = \frac{F_C}{U} = \frac{F_{C0}[\Theta_C + (c/a)X]}{U} = \frac{F_{C0}[\Theta_C + (c/a)X]}{U_0} = C_{C0}[\Theta_C + (c/a)X]$$

$$C_D = \frac{F_D}{U} = \frac{F_{D0}[\Theta_D + (d/a)X]}{U} = \frac{F_{D0}[\Theta_D + (d/a)X]}{U_0} = C_{D0}[\Theta_D + (d/a)X]$$

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CONCENTRATION IN TERMS OF CONVERSION

2. For gas phase:

- Batch System

$$C_A = \frac{N_A}{V} \quad \text{Need to substitute } V \text{ from gas law equation}$$

From equation of state:

At any time t ,

$$PV = ZN_T RT \quad (1)$$

At initial condition ($t=0$)

$$P_0 V_0 = Z_0 N_{T0} RT_0 \quad (2)$$

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CONCENTRATION IN TERMS OF CONVERSION

For gas phase:

- Batch System

$$V = V_0 \left(\frac{P_0}{P} \right) \frac{T}{T_0} \left(\frac{Z}{Z_0} \right) \frac{N_T}{N_{T0}} \quad (3)$$

Substituting the expression for N_T/N_{T0} in (3),

$$V = V_0 \left(\frac{P_0}{P} \right) \frac{T}{T_0} \left(\frac{Z}{Z_0} \right) (1 + \epsilon X) \quad \text{If the compressibility factor is not change significantly during rxn, } Z_0 \approx Z$$

$$V = V_0 \left(\frac{P_0}{P} \right) (1 + \epsilon X) \frac{T}{T_0} \quad (5)$$

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CONCENTRATION IN TERMS OF CONVERSION

2. For gas phase:

- Batch System

$$\frac{N_T}{N_{T0}} = 1 + \delta \sum_{j=1}^J \nu_j X$$

$$\frac{N_T}{N_{T0}} = 1 + \epsilon X$$

Rearranging:

$$\epsilon = \frac{N_T - N_{T0}}{N_{T0} X}$$

At complete conversion (for irreversible rxn): $X=1$, $N_T=N_{Tf}$

$$\epsilon = \frac{N_{Tf} - N_{T0}}{N_{T0}}$$

 ϵ is the fractional change in volume of the system between no conversion and complete conversion of reactant A

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CONCENTRATION IN TERMS OF CONVERSION

For gas phase:

• Flow System

$$C_j = \frac{F_j}{v} \quad \text{Need to substitute } v \text{ from gas law equation}$$

From gas law, at any point in the reactor,

$$C_j = \frac{F_j}{v} = \frac{P}{ZRT} \quad (1)$$

At the entrance of reactor:

$$C_{j0} = \frac{F_{j0}}{v_0} = \frac{P_0}{Z_0 R T_0} \quad (2)$$

$$v = v_0 \left(\frac{F_T}{F_{T0}} \right) \left(\frac{P_0}{P} \right) \left(\frac{T}{T_0} \right) \quad (3) \quad \text{Dividing (1) by (2)}$$

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CONCENTRATION IN TERMS OF CONVERSION

For gas phase:

• Flow System

Recall from stoichiometric table

$$F_T = F_{T0} + F_{A0} \delta X$$

$$v = v_0 \left(\frac{F_T}{F_{T0}} \right) \left(\frac{P_0}{P} \right) \left(\frac{T}{T_0} \right)$$

Substituting for F_T :

$$\begin{aligned} v &= v_0 \left(\frac{F_{T0} + F_{A0} \delta X}{F_{T0}} \right) \left(\frac{P_0}{P} \right) \left(\frac{T}{T_0} \right) \\ &= v_0 \left(1 + \frac{F_{A0}}{F_{T0}} \delta X \right) \left(\frac{P_0}{P} \right) \left(\frac{T}{T_0} \right) \\ &= v_0 \left(1 + y_{A0} \delta X \right) \left(\frac{P_0}{P} \right) \left(\frac{T}{T_0} \right) \\ &= v_0 (1 + \epsilon X) \left(\frac{P_0}{P} \right) \left(\frac{T}{T_0} \right) \quad (4) \end{aligned}$$

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CONCENTRATION IN TERMS OF CONVERSION

For gas phase:

• Flow System

$$C_j = \frac{F_j}{v} \quad \text{Need to substitute } v \text{ from gas law equation}$$

$$v = v_0 (1 + \epsilon X) \left(\frac{P_0}{P} \right) \left(\frac{T}{T_0} \right) \quad (4) \quad F_j = F_{j0} (\Theta_j + C_j X) \quad (5)$$

Substituting v & F_j :

$$\begin{aligned} C_j &= \frac{F_{j0} (\Theta_j + C_j X)}{v_0 (1 + \epsilon X) \left(\frac{P_0}{P} \right) \left(\frac{T}{T_0} \right)} \\ &= C_{j0} \frac{(\Theta_j + C_j X)}{(1 + \epsilon X)} \left(\frac{P}{P_0} \right) \left(\frac{T_0}{T} \right) \end{aligned}$$

Stoichiometric coefficient
(d/a , c/a , $-b/a$, $-a$)

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CHE 7 311 Kinetics & Reactor Design

Flow System

Concentration for each species:

$$\begin{aligned} C_A &= \frac{F_A}{v} = \frac{F_{A0}(1-X)}{v} = C_{A0} \left(\frac{1-X}{1+\epsilon X} \right) \left(\frac{T_0}{T} \right) \left(\frac{P}{P_0} \right) \quad v = v_0 (1 + \epsilon X) \left(\frac{P_0}{P} \right) \left(\frac{T}{T_0} \right) \\ C_B &= \frac{F_B}{v} = \frac{F_{B0} - (b/a)X}{v} = C_{B0} \frac{(\Theta_B - (b/a)X)}{(1 + \epsilon X)} \left(\frac{T_0}{T} \right) \left(\frac{P}{P_0} \right) \\ C_C &= \frac{F_C}{v} = \frac{F_{C0} + (c/a)X}{v} = C_{C0} \frac{(\Theta_C + (c/a)X)}{(1 + \epsilon X)} \left(\frac{T_0}{T} \right) \left(\frac{P}{P_0} \right) \\ C_D &= \frac{F_D}{v} = \frac{F_{D0} + (d/a)X}{v} = C_{D0} \frac{(\Theta_D + (d/a)X)}{(1 + \epsilon X)} \left(\frac{T_0}{T} \right) \left(\frac{P}{P_0} \right) \\ C_I &= \frac{F_I}{v} = \frac{F_{I0} \Theta_I}{v} = \frac{C_{I0} \Theta_I}{(1 + \epsilon X)} \left(\frac{T_0}{T} \right) \left(\frac{P}{P_0} \right) \end{aligned}$$

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CHE 7 511 Kinetics & Reactor Design



- Milk is pasteurized if it is heated to 63°C for 30 min, but if it is heated to 74°C it only needs 15 s for the same result. Find the activation energy of this sterilization process.

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Example

- Pure gaseous reactant A ($CA_0 = 100$ millimol/liter) is fed at a steady rate into a mixed flow reactor ($V = 0.1$ liter) where it dimerizes ($2A \rightarrow R$). For different gas feed rates the following data are obtained. Find a rate equation for this reaction

Run number	1	2	3	4
v_0 , liter/hr	10.0	3.0	1.2	0.5
C_{A1} , millimol/liter	85.7	66.7	50	33.4

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CHE 7 511 Kinetics & Reactor Design



Example

- The homogeneous gas decomposition of phosphine $4 PH_3(g) \rightarrow P_4(g) + 6 H_2(g)$ proceeds at 649 °C with the first-order rate $-r_{PH_3} = \left(\frac{10}{\text{hr}}\right) C_{PH_3}$. What size of plug flow reactor operating at 649°C and 460 kPa can produce 80% conversion of a feed consisting of 40 mol of pure phosphine per hour?



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